Edge-Efficient Deepfake Detection via Hierarchical Temporal Modeling

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Project Repository

Abstract

The widespread use of social media has led to an unprecedented volume of video content, making large-scale deepfake detection computationally challenging. Running heavy detection models on every video is impractical due to latency, energy, and resource constraints, especially on low-compute platforms such as mobile phones, embedded systems, and edge devices. However, most state-of-the-art deepfake detectors rely on large neural networks, limiting their real-time applicability in such settings. In this work, we propose an efficient video deepfake detection framework based on a MobileNet backbone augmented with Temporal Shift Modules (TSM) and a Hierarchical TSM (HTSM) for multi-scale temporal modeling. Our approach captures both local and global temporal dependencies using sparsely sampled frames while maintaining low computational overhead. We evaluate our method on FaceForensics++ and Celeb-DF using performance metrics and edge-centric metrics. Experimental results show that our method achieves competitive detection performance while significantly reducing computational cost, making it well-suited for real-time deepfake detection on resource-constrained devices.

1. Introduction

Deepfake detection has become an important problem in computer vision and multimedia forensics due to rapid advancements of models that produce highly realistic synthetic videos. Early methods focused on detecting spatial artifacts within individual frames, such as inconsistencies introduced during face warping or blending [1, 7]. Subsequent approaches incorporated temporal modeling, using recurrent networks and spatiotemporal architectures to capture motion inconsistencies across frames [5, 11]. Recently, transformer-based and hybrid architectures have achieved strong performance by modeling complex spatial and temporal dependencies [3, 12]. However, these methods typically rely on high-capacity models with significant computational and memory requirements, limiting their applicability in real-time or resource-constrained environments.

At the same time, large-scale video platforms require efficient moderation tools, where processing every uploaded video using heavy models is impractical due to latency, energy, and deployment constraints. This has motivated research into lightweight deepfake detection methods. Prior work explores compact architectures such as EfficientNet variants, CNN-LSTM hybrids, and other efficiency-driven designs [2, 4]. While these approaches reduce computational cost, they often introduce trade-offs in temporal modeling capability, scalability, or generalization across datasets.

In contrast, efficient temporal modeling techniques such as the Temporal Shift Module (TSM) demonstrate that temporal reasoning can be incorporated into convolutional networks with negligible computational overhead [9]. Similarly, lightweight backbone architectures such as MobileNet enable efficient feature extraction suitable for edge devices [6]. Building on these ideas, we propose an edge-efficient deepfake detection framework based on a MobileNet backbone augmented with TSM and a Hierarchical Temporal Shift Module (HTSM). The proposed HTSM captures both intra-region (local) and inter-region (global) temporal dependencies over sparsely sampled frames, enabling multi-scale temporal aggregation under strict computational budgets.

We evaluate our approach on widely used benchmark datasets, FaceForensics++ [10] and Celeb-DF [8], using both predictive metrics (ROC-AUC, PR-AUC, F1, MCC, and balanced accuracy) and system-level metrics (model size, latency, energy consumption, and CO_2 emissions). Experimental results show that our method achieves competitive performance while significantly improving efficiency, and demonstrates better cross-dataset generalization compared to baseline lightweight models. These results highlight the practicality of our approach for real-world deployment on edge devices.

Contributions:

1. We propose an edge-efficient video deepfake detection framework based on a MobileNet backbone augmented with Temporal Shift Modules (TSM), enabling temporal modeling with negligible computational overhead.

2. We introduce a Hierarchical Temporal Shift Module (HTSM) that performs multi-scale temporal aggregation by modeling both intra-region (local) and inter-region (global) dependencies across sparsely sampled frames.
3. We conduct diverse experiments evaluating both predictive performance and edge-centric efficiency, including analysis across multiple metrics, performance scaling with training data size, and cross-dataset generalization through cross-data evaluation and model merging.

2. Related Works

2.1. Deepfake Detection

Deepfake detection has been extensively studied using spatial, temporal, and hybrid modeling approaches. Early methods primarily relied on convolutional neural networks (CNNs) to detect spatial inconsistencies such as face-warping artifacts and mesoscopic image features [1, 7]. While effective, these approaches operate at the frame level and often fail to capture temporal inconsistencies present in manipulated videos.

To address this limitation, subsequent works incorporated temporal modeling by combining CNN-based feature extraction with recurrent architectures such as LSTMs [11]. Similarly, spatiotemporal CNNs were proposed to jointly model spatial and temporal information, improving robustness to video-level manipulations [5]. Hybrid CNN-LSTM architectures further became a common paradigm for capturing short-term temporal dependencies.

More recently, high-capacity models such as transformer-based and hybrid architectures have achieved strong performance by modeling long-range dependencies and multi-scale features [3, 12]. These approaches are effective across diverse datasets and manipulation techniques but typically incur significant computational and memory overhead. As a result, their deployment in real-time or resource-constrained environments remains challenging.

2.2. Deepfake Detection on Edge Devices

To mitigate the computational cost of deepfake detection, recent work has explored lightweight and efficiency-oriented approaches. One direction focuses on designing compact architectures, such as EfficientNet-based models and hybrid CNN-LSTM frameworks, which aim to balance performance and efficiency [2, 4]. While these methods reduce model complexity, they often introduce trade-offs in temporal modeling capability or generalization.

Another line of work emphasizes reducing inference cost through input-level optimizations, such as frame sampling and region-based processing. By limiting the number of processed frames or focusing on facial regions, these approaches reduce computational overhead while maintaining reasonable detection performance.

In contrast, efficient temporal modeling techniques such as the Temporal Shift Module (TSM) enable temporal reasoning within standard CNNs at negligible computational cost [9]. Combined with lightweight backbone networks such as MobileNet [6], these approaches provide a promising direction for edge deployment. Building on this idea, our work integrates a MobileNet backbone with TSM and extends it through a Hierarchical Temporal Shift Module (HTSM) to capture both local and global temporal dependencies under strict computational constraints. Additionally, we incorporate an improved frame sampling strategy to maximize temporal coverage while maintaining efficiency.

3. Methodology

3.1. Overview

We propose a family of lightweight architectures for video deepfake detection designed for low-compute environments. The framework is built on a MobileNet backbone for efficient spatial feature extraction, combined with lightweight temporal modeling modules to capture inter-frame dependencies.

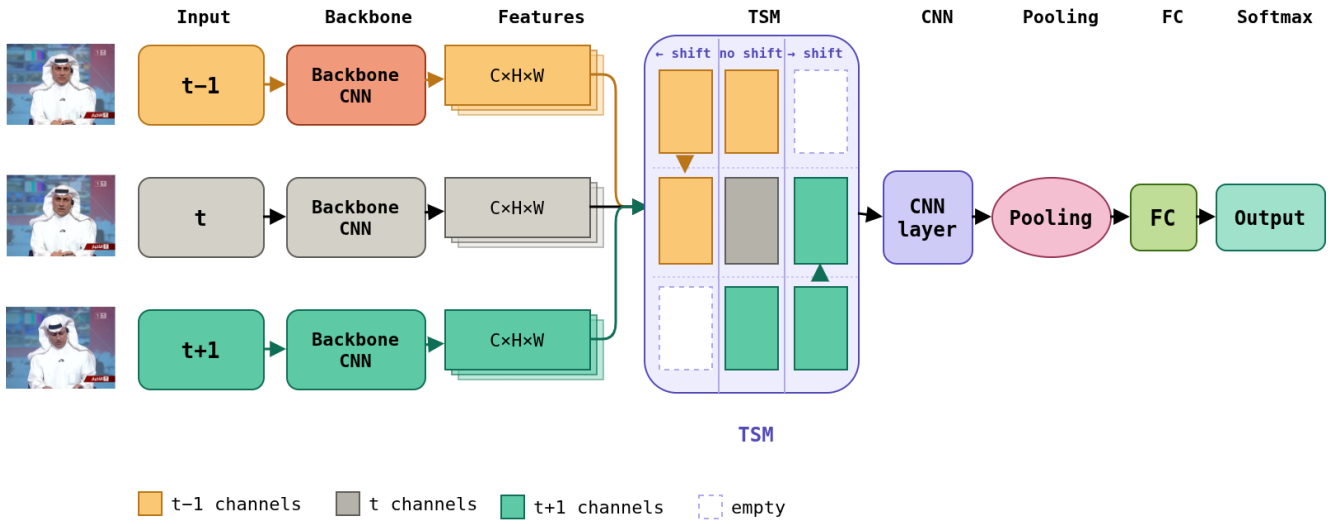
Given a sequence of sampled frames, each frame is processed independently to extract spatial features. Frames are sparsely sampled from each video; we defer detailed sampling and preprocessing specifics to the experimental setup.

3.2. MobileNet + TSM

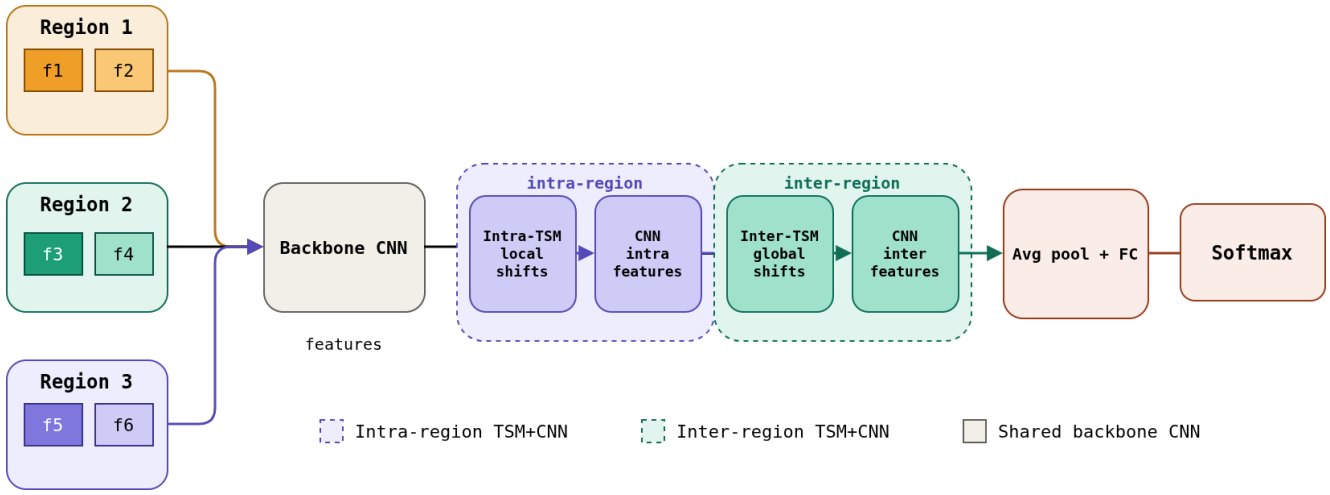
Our baseline model integrates a Temporal Shift Module (TSM) with a MobileNet backbone. Given a sequence of sampled frames, each frame is processed independently to extract spatial features. The TSM then enables temporal interaction by shifting a subset of feature channels across adjacent frames, introducing temporal context at negligible computational cost. The temporally enhanced features are refined using a lightweight convolutional layer, followed by channel-wise global average pooling. Temporal aggregation is performed simple averaging across frames to obtain a video-level representation, which is passed through a lightweight fully connected classifier.

3.3. MobileNet + Hierarchical TSM (HTSM)

To improve temporal modeling capacity, we introduce a Hierarchical Temporal Shift Module (HTSM) that captures both local and global temporal dependencies. The architecture consists of two sequential stages: an intra-region TSM followed by an inter-region TSM. The intra-region module models short-range temporal interactions within local frame groups, while the inter-region module captures longer-range dependencies across distant frames. Each stage is followed by a lightweight convolutional layer for feature refinement. The temporally enhanced features are then subjected to channel-wise global average pooling. Temporal aggrega-



(a) Temporal Shift Module (TSM)



(b) Hierarchical Temporal Shift Module (HTSM)

Figure 1. Comparison of temporal modeling architectures. TSM performs channel-wise temporal shifts across adjacent frames, while HTSM extends this with hierarchical temporal aggregation for richer spatiotemporal representation.

tion is performed via simple averaging across frames to obtain a video-level representation, which is passed through a lightweight fully connected classifier.

3.4. MobileNet + Residual HTSM

We further extend HTSM by introducing residual connections across temporal modeling blocks. Specifically, skip connections are added over the intra-TSM and inter-TSM stages to facilitate feature reuse and improve gradient flow. This design choice is inspired by the original TSM formulation, where residual connections are recommended to preserve spatial feature integrity during temporal shifting [9]. As a result, the proposed design helps preserve information across transformations and stabilizes training, while incurring minimal additional cost.

4. Experiments & Results

4.1. Experimental Setup

We evaluate our proposed architectures on two widely used deepfake detection benchmarks: FaceForensics++ (FF++) [10] and Celeb-DF [8], which provide diverse and realistic manipulation scenarios for robust evaluation. The dataset composition is summarized in Table 4.

Table 4. Dataset composition of the evaluation benchmarks.

Dataset	Real Videos	Manipulated Videos
FaceForensics++ (FF++)	1,000	5,000
Celeb-DF	890	5,639

Table 1. Performance comparison of models on FF++ and Celeb-DF datasets.

Model	FF++					Celeb-DF				
	ROC	PR	F1	MCC	BalAcc	ROC	PR	F1	MCC	BalAcc
XceptionNet	**0.9911**	**0.9967**	**0.9552**	**0.9109**	**0.9620**	0.9963	0.9985	0.9680	0.9366	0.9610
MobileNet	0.9808	0.9898	0.9422	0.8868	0.9290	0.9962	0.9985	0.9686	0.9372	0.9700
MobileNet + TSM	0.9799	0.9895	0.9407	0.8814	0.9420	**0.9964**	**0.9986**	**0.9756**	**0.9512**	**0.9770**
MobileNet + HTSM	0.9865	0.9932	0.9453	0.8948	0.9280	0.9928	0.9969	0.9684	0.9368	0.9670
MobileNet + Residual HTSM	0.9867	0.9929	0.9453	0.8920	0.9560	0.9927	0.9970	0.9585	0.9173	0.9640

Table 2. Edge-device performance (CPU inference).

Model	Size (MB)	Lat. (ms)	Energy (nWh)	CO_2 (μg)
XceptionNet	83.7	2601	8854	34.9
MobileNet	42.4	234	819	3.23
MobileNet + TSM	42.4	245	810	3.19
MobileNet + HTSM	78.5	250	845	3.33
MobileNet + Residual HTSM	78.5	247	865	3.40

To ensure computational efficiency, all models operate on sparsely sampled video frames. We sample a total of 20 frames per video, organized into 4 uniformly separated temporal regions with 5 consecutive frames per region, enabling both local and global temporal coverage. The data processing pipeline consists of frame sampling followed by face cropping, resulting in input tensors of size $N \times 20 \times 3 \times 224 \times 224$. In addition, we perform experiments across varying training data sizes and conduct cross-dataset evaluation, including model merging, to assess generalization.

4.2. Evaluation Metrics

We evaluate all models using a comprehensive set of predictive and efficiency metrics. For predictive performance, we report ROC-AUC and PR-AUC, F1-score, Matthews Correlation Coefficient (MCC), and balanced accuracy to provide a robust assessment under class imbalance.

To assess deployment feasibility on edge devices, we perform a standardized evaluation of system-level efficiency. Specifically, we measure inference latency, energy consumption using CodeCarbon (kWh and CO_2 equivalent emissions), and model size in terms of disk footprint. The evaluation was conducted on a video of length 15 seconds.

The evaluation follows a strict protocol to ensure consistency and reproducibility. All experiments are performed on CPU-only execution in a single-threaded setting, with GPU acceleration and mixed precision disabled. Models are evaluated in inference mode using `torch.no_grad()` and `eval()` settings. Warm-up runs are excluded from timing, and latency is computed over multiple runs to report mean and standard deviation.

4.3. Baselines

First, we use a frame-based MobileNet model without temporal modeling as a lightweight baseline. This allows us to evaluate the impact of introducing temporal modules over a purely spatial architecture. Second, we include XceptionNet as a high-capacity baseline, which is widely used in deepfake detection and serves as a strong reference for performance.

This comparison enables us to analyze the trade-off between computational efficiency and detection performance, and to quantify the benefits of incorporating efficient temporal modeling within a lightweight framework.

4.4. Effect of Hierarchical Temporal Modeling

To evaluate the effectiveness of the proposed hierarchical design, we replace the hierarchical sampling strategy with uniform sampling while keeping the model architecture unchanged. As shown in Table 3(a), this results in a noticeable drop in performance, indicating that the hierarchical formulation both in terms of temporal modeling and frame selection plays a critical role in capturing informative temporal dynamics.

4.5. Effect of Training Data Size

Due to computational constraints, we train our models on varying fractions of the FF++ training set:

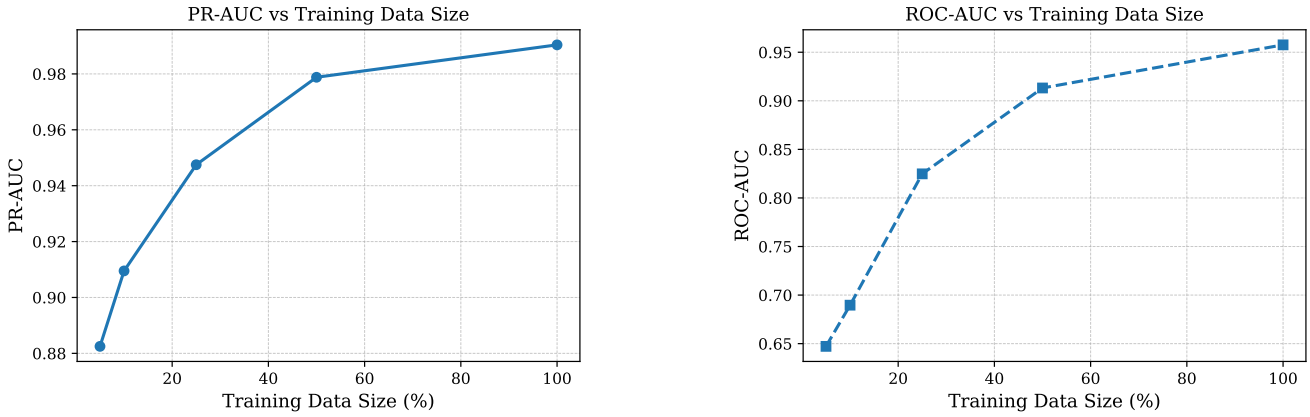
Table 3. Performance comparison and cross-dataset generalization results.

(a) Sampler Comparison

Model	PR-AUC	ROC-AUC
MobileNet + Residual HTSM + Uniform Sampler	0.9893	0.9545
MobileNet + Residual HTSM + Our Sampler	**0.9905**	**0.9601**

(b) Cross-Dataset Generalization

Train → Test	ROC-AUC	PR-AUC
FF++ → FF++	**0.9867**	**0.9929**
FF++ → Celeb-DF	0.7298	0.8687
Celeb-DF → FF++	0.6354	0.8183
Celeb-DF → Celeb-DF	**0.9927**	**0.9970**
Merged → FF++	0.8568	0.9387
Merged → Celeb-DF	0.9524	0.9701



(a) PR-AUC (b) ROC-AUC

Figure 2. Performance as a function of training data size. Both PR-AUC and ROC-AUC improve consistently with increasing data, demonstrating strong scaling behavior under constrained training settings.

$5\%, 10\%, 25\%, 50\%, 100\%$, and evaluate performance using ROC-AUC and PR-AUC.

As shown in Fig. 2, both ROC-AUC and PR-AUC improve consistently with increasing training data size, indicating a clear positive correlation between data availability and detection performance. The monotonic trend suggests that the model has not yet saturated and can benefit from additional data.

4.6. Cross-Dataset Evaluation

To evaluate generalization, we train models on one dataset and test on another (FF++ → Celeb-DF and vice versa), reflecting real-world scenarios where models encounter unseen manipulation distributions.

As shown in Table 3 (b), there is a clear drop in performance under cross-dataset evaluation compared to in-domain results, highlighting the presence of a significant domain gap. Models trained on one dataset struggle to generalize effectively to another due to differences in data distribution and manipulation characteristics.

W also evaluate a merged training setup, which consistently improves cross-dataset performance, demonstrating that exposure to more diverse training data enhances generalization. Overall, these results indicate that while domain shift remains a key challenge, incorporating diverse data and effective temporal modeling contributes to improved robustness across datasets.

4.7. Key Observations

Table 1 presents the overall performance comparison across all models. We observe two notable trends across datasets. On FF++, the HTSM variant performs better than its residual counterpart, which we attribute to the lack of fine-tuning of residual connections, leading to suboptimal feature integration. In contrast, on Celeb-DF, HTSM underperforms compared to simpler variants. This can be attributed to the limited temporal context provided by the sampled 20 frames, which may introduce noise rather than useful temporal signals. Increasing the temporal context through more extensive frame sampling could potentially improve performance in such cases.

Overall, these findings demonstrate that while hierarchical temporal modeling is beneficial, its effectiveness depends on appropriate temporal coverage and careful architectural tuning.

5. Limitations and Future Work

Generalization Across Datasets. Our models are trained on FaceForensics++ and Celeb-DF and exhibit performance degradation under cross-dataset evaluation, highlighting limited generalization to unseen data distributions. Improving robustness through larger and more diverse training data remains an important direction.

Limited Training Scale. Due to computational constraints, we rely on pre-trained MobileNet features and do not perform full end-to-end training. This may limit the model's ability to learn deepfake-specific representations. Future work could explore large-scale training and task-specific pretraining.

Architectural and Hyperparameter Constraints. Our exploration of architectural design and hyperparameters is limited. In particular, integrating TSM within backbone layers and performing systematic tuning of temporal modeling components could further improve performance.

Limited Comparison with Recurrent Temporal Models. Our work focuses on efficient TSM-based temporal modeling and does not compare with recurrent methods (e.g., GRU, LSTM), which, while effective, incur higher computational cost. A systematic comparison is left for future work.

Temporal Sampling Strategy. Our current frame sampling is simple and may miss informative temporal segments. Adaptive or content-aware sampling, along with increased temporal context via more frames, could improve performance.

Robustness to Advanced Generative Models. Recent generative models, especially diffusion-based methods, produce highly realistic outputs with fewer detectable artifacts. Adapting detection methods to handle such data remains an open challenge.

Feature Representation. Our framework relies on RGB-based spatial and temporal features. Incorporating complementary cues such as frequency-domain or multi-modal features could improve robustness without significantly increasing computational cost.

Deployment Scope. Although we evaluate efficiency on a CPU setup, experiments are limited to a single hardware configuration. Broader evaluation across diverse edge devices would strengthen claims of real-world applicability.

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Hyperparameters:

Training hyperparameters used for all experiments:

- Optimizer: AdamW
- Learning rate: $1e{-5}$
- Weight decay: $5e{-4}$
- Loss: BCEWithLogitsLoss
- Precision: AMP
- Sampler: WeightedRandomSampler
- Early stopping: PR-AUC
- Data split: Train / Validation / Test = 75% / 15% / 10%